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MILITARY SPECIFICATION

POWER SUPPLY, METALLIC RECTIFIER TYPE GENERAL SPECIFICATION FOR

* This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general requirements for a portable or stationary metallic rectifier power supply.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

* SPECIFICATIONS

Federal

O-E-910	Extinguisher, Fire, Carbon dioxide (Hand and Wheel Types)

Military

MIL-P-116	Preservation-Packaging, Methods Of
MIL-C-5756	Cable and Wire, Power, Electric, Portable
MIL-N-7513	Nomenclature Assignment, Contractors Method for Obtaining
MIL-A-8421	Air Transportability Requirements, General Specification for

STANDARDS

* Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

FSC 6130

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MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipment)
MIL-STD-756	Reliability Prediction
MIL-STD-794	Part and Equipment, Procedures for Packaging and Packing of
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment: Requirements for
MIL-STD-889	Dissimilar Metals

HANDBOOK

AFSC Design Handbook

DH 1-6	Air Force System Command Design Handbook-System Safety
DH 1-X	Air Force System Command Design Handbook-Checklist of General Design Criteria

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction testing. This specification makes provisions for preproduction testing.

3.2 Components. The power supply shall consist of the following major components:

Item Name	See Requirement
Rectifier elements	3.7.1
Transformer	3.7.2
Power cables	3.7.3
Auxiliary equipment	3.7.4
Cabinet	3.7.5.

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3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Fungus-proof material. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent. However, if they will be used in a hermetically-sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.4.2.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.5 Design and construction. The power supply shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5.1 Adjustments and repairs. The power supply shall be so constructed that adjustments and repairs can be easily made by the personnel of operational units and overhaul bases.

3.5.2 Simplification. The same part shall be used for as many applications as practicable in a given power supply and in power supplies of different ratings.

3.5.3 Reliability. The power supply shall have a predicted MTBF (mean-time-between-failures) of not less than 2,000 hours. The prediction shall conform to the following conditions:

a. For reliability purposes, a failure shall be any failure or malfunction that could prevent safe operation or that could prevent satisfactory operation in the primary functions of supplying rectified power when demanded. Malfunctions that would not prevent such performance, such as loose bolts in noncritical applications, and preventive or scheduled maintenance to preclude wear-out or time dependent failures shall not be considered reliability failures

b. The prediction shall be based on the principles of MIL-STD-756, Reliability Prediction, or the assumption that the random failure rate of the system can be estimated as a function of component random failure histories and estimates.

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Failure rate data and estimates used in the prediction shall be modified to account for component use times and for stresses, loads, and environments expected compared to those of the data

c. The intent of the prediction shall be not to require extensive part testing to develop component failure rates. The intent of the prediction shall be to apply each component based on engineering evaluation of failure frequency data and observations of identical or similar components and designs in related environments and applications. The contractor shall have available to him documented failure rate data and estimates used as a basis for the prediction.

3.5.4 Maintainability. The power supply shall be so designed and constructed that rectifiers, transformers, relays, switches, circuit breakers, meters, and other parts for which replacement may be economically practicable can be readily replaced at maintenance levels lower than depot using only common handtools and equipment commonly expected in shops doing maintenance on similar equipment.

3.5.4.1 The design and construction shall be in accordance with the design guidelines of MIL-STD-470, paragraph titled, Establish Maintainability Design Criteria, except beyond being considered, the guidelines shall be applied wherever appropriate to the subject class or type of item and where the guidelines do not conflict with other requirements specified herein. Repairable printed circuit assemblies where used, shall include test points that are labeled, color coded and in sequential order.

3.5.4.2 Active maintenance downtime. The active maintenance downtime for each maintenance action, including corrective and preventive maintenance, servicing, and adjustments, expected or planned below depot level maintenance shall be not more than 20 minutes. Protective coatings where used shall be transparent and capable of being removed with a low wattage iron.

3.5.5 Rated output and overload. The power supply shall provide continuous output rating when the input voltage is within ±10 percent of rated input as specified in the detail specification. The power supply shall have an intermittent overload capacity of 25 percent of rated load (see 6.3.1).

3.5.6 The input voltage, voltage regulation, ripple voltage, and efficiency of the power supply shall be as specified in the detail specification.

3.5.6.1 The regulated voltage band shall be as defined in the detail specification under the paragraph entitled, Regulation.

3.5.7 Stability. Thr portable power supply shall be so designed that it can be tilted 30° from the normal position and will return to the normal stationary position upon release.

3.5.8 Air transportability. The power supply shall be designed to meet the requirements of MIL-A-8421.

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* 3.5.9 Fire extinguisher. If required by the detail specification, a 15-pound carbon dioxide fire extinguisher in accordance with O-E-910 shall be installed in a quick-release-type bracket in a readily accessible location.

* 3.5.10 System safety. The power supply shall be designed using the applicable system safety design criteria provided in AFSC DH 1-6, Chapter 4, and AFSC DH 1-X, Chapter 6. The power supply shall be so designed that the design will not result in a Category III or Category IV hazard as defined in MIL-STD-882, for the predicted operating life of the equipment.

* 3.5.11 Personnel safety. The power supply shall be designed in accordance with MIL-STD-454, Requirement 1.

3.6 Performance. The power supply shall withstand the following conditions:

a. Operating temperatures ranging from -46° to +55°C

b. Storage temperatures ranging from -46° to +71°C

c. Radiant energy as found under natural conditions

d. Rainfall of 4 ±1 inch per hour

e. Relative humidity up to 100 percent

f. Fungus growth as encountered in tropical climates

g. Exposure to salt fog

h. Sand and dust particles as encountered in desert areas.

3.6.1 Parallel operation. The power supply shall be capable of parallel operation with any other power supply of the same rating and manufacturer's part number.

3.6.2 Dielectric. The power supply shall withstand twice rated input voltage +1,000V rms, 60 cycles, between the primary winding and the core and the secondary winding, and 500V between the secondary winding and the core.

3.6.3 Surge rating. The power supply shall be capable of carrying 400 percent rated current for 3 seconds at three surges per minute for 30 minutes.

3.6.4 Transient load response. The output of the power supply shall not vary from the regulated voltage band by more than 25 percent when a transient load equal to the intermittent overload capacity is applied or removed, thus causing the power supply output to change from no load to overload and vice versa. The voltage shall recover to, and remain within, ±5 percent of the regulated voltage

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band in 0.2 second, or less, and shall recover to, and remain within ±2 percent of the regulated band in 0.4 second, or less.

3.6.5 Electrical connections

3.6.5.1 Current regulation. The current regulation system shall hold the output current within ±10 percent of any preselected value which shall be adjustable as specified in the detail specification. The current regulation system shall operate over an output range of 6V to a preselected maximum value which shall be adjustable from 28 to 40V and in no case shall exceed 40V.

3.6.5.2 Single rectifier set operation. The electrical connections shall be arranged to permit the regulated voltage output of the power supply to be delivered through both cables from the time the power supply is energized until the jet engine starter control switch is operated or the signal is applied. The power supply output voltage characteristics shall change to provide constant current to that cable attached to the constant current receptacle on the aircraft while continuing to provide regulated voltage to the remaining cable. The power supply shall be capable of supplying the constant current output to either of the output cables. When the power supply is operating under split-bus conditions, the portion of the power supply furnishing constant current shall be electrically isolated from the portion supplying regulated voltage.

3.6.5.3 Dual power supply operation. When two power supplies are operating in parallel and their output cables are connected to the aircraft external power receptacles, the electrical connections shall permit the regulated voltage outputs of the power supplies to be delivered through both cables. The output characteristics of the power supply receiving the signal shall change to provide constant current to that cable attached to the constant-current receptacles on the aircraft. The power supply not receiving the signal shall continue to provide regulated voltage output to the aircraft. The system shall be capable of supplying the constant current output to either of the output cables.

3.7 Details of components

3.7.1 Rectifier elements. Rectifier elements shall be of the metallic type.

3.7.2 Transformer. The transformer shall be of the insulated type having separate high- and low-voltage windings.

3.7.2.1 Iron core. The iron core of the transformer shall be securely clamped and bolted to eliminate shifting or loosening of the laminations due to shock, stresses, vibration, and other conditions encountered during service, operation, shipping, and storage.

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3.7.2.2 Insulation. The transformer assembly shall be impregnated with a good quality insulating compound and oven-baked to secure proper drying in accordance with National Electrical Manufacturers Association's standards for this class of equipment.

3.7.2.3 Temperature rise. The transformer temperature rise while operating at continuous rating shall not exceed 50°C for class A insulation, 75°C for class B insulation, nor 120°C for class H insulation.

3.7.2.4 Wedges. The transformer wedges shall be of hard insulating material impervious to heat and moisture and shall hold the windings securely throughout the life of the transformer.

3.7.3 Power cables

3.7.3.1 Input cable. The input cable shall be a 600-V, extra-flexible, heavy-duty, portable, rubber or synthetic-rubber-covered cable and shall be furnished in either of two types as specified by the procuring activity (see 6.2). Type I shall be in accordance with MIL-C-5756 and shall be identified by a permanent yellow band adjacent to each end. Type II cable shall be similar to type I, except it shall be high-quality commercial cable which need not be flexible at low temperatures. The allowable voltage drop or cable size and length of the power supply cable shall be as specified in the detail specification.

3.7.3.2 Output cable. The output cable shall conform to the requirements specified in the detail specification.

3.7.4 Auxiliary equipment. Auxiliary equipment shall be provided as follows:

3.7.4.1 Starting switch and overload relays. A starting switch and overload relays shall be incorporated in the input circuit to provide protection against prolonged overloads. The overload relays shall permit continuous operation at 100 percent of rated output at an ambient temperature of not less than +35°C.

3.7.4.2 Voltmeter. A voltmeter shall be provided and shall have an accuracy of ±2 percent, a dial size not less than 3-1/2 inches, and a scale range as specified in the detail specification.

3.7.4.3 Ammeter. An ammeter shall be provided and shall have a accuracy of ±2 percent, a dial size not less than 3-1/2 inches, and a scale range as specified in the detail specification.

3.7.4.4 Shorting switch. A shorting switch (normally closed) shall be provided for the ammeter circuit. When the switch is in the closed position, there shall be no deflection of the ammeter at full power supply load.

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3.7.4.5 Pilot light. A red pilot light shall be provided to indicate when the power supply is operating.

3.7.4.6 Fan motor. The fan motor shall have a minimum lubricating period of 6 months and shall be baffled sufficiently to withstand the design operating conditions specified herein. The fan motor shall incorporate a means to stop operation of the power supply in the event of failure of the ventilating systems.

3.7.5 Cabinet. The cabinet shall consist of a weatherproof, sheet metal enclosure containing the power supply and all auxiliary equipment. A watertight cable entrance and bushings shall be installed in the cabinet for each cable.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

* 3.9 Electromagnetic interference. The power supply shall comply with the requirements of MIL-STD-461 and MIL-STD-462.

3.10 Dimensions. The dimensions of the power supply shall be as specified in the detail specification.

3.11 Weight. The weight of the power supply shall be as specified in the detail specification.

3.12 Finishes and protective coatings. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808. Exterior surfaces of the power supply shall be finished in yellow, film designation DG, and interior surfaces shall be gray, film designation LG.

3.13 *Operational markings*

3.13.1 Operating instructions. Permanent visual operating instructions shall be provided on the front panel containing a statement as to the correct adjustment for the operating input-voltage range.

3.13.2 Circuit diagram. A wiring diagram shall be permanently affixed to the back of the removable or hinged panel. The diagram shall be clear, readable, and weather resistant.

3.14 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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3.14.1 Type designation. The power supply type designation specified in the detail specification shall be placed on the front and rear of the unit. The letters shall be black and shall be 4 inches high with 1/2-inch line width. Nomenclature assignment and nameplate approval shall be obtained as specified in MIL-N-7513.

3.15 Workmanship. The power supply, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; and alignment of parts and tightness of assembly screws, bolts, et cetera.

3.15.1 Riveting. Riveting operations shall be performed to insure that the rivets are tight and satisfactorily headed.

3.15.2 Cleaning. The power supply shall be thoroughly cleaned, and loose, spattered, or excess solder; metal chips; and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the power supply shall be as follows:

a. Preproduction testing . . . . . See 4.4

* b. Quality conformance testing . . . . . See 4.5.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

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4.4 Preproduction testing (see 6.2)

4.4.1 Test samples. Two power supplies shall be subjected to the preproduction tests specified in 4.4.3.

4.4.2 Test report. After completion of the preproduction test, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

* 4.4.2.2 System safety. Identification of system hazards shall be accomplished by conducting the following analysis as outlined in MIL-STD-882 paragraph entitled, System Hazard Analyses. The results of these Hazard Analyses shall be included in the applicable operating documents.

a. A Subsystem Hazard Analysis shall be accomplished in accordance with the Fault Hazard Analysis in AFSC DH 1-6, Chapter 2.

b. An operating Hazard Analysis shall be accomplished.

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* 4.4.2.3 Personnel safety. The power supply shall be tested in accordance with MIL-STD-454 Requirement 1.

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests specified under 4.6. Unless otherwise specified, the test samples shall be allocated to the preproduction tests as follows and as described under 4.6:

Sample No. 1	Sample No. 2
Examination of product	Examination of product
Dielectric	Dielectric
Operational	Operational
Regulation	Regulation
Efficiency	Efficiency
Stability	Parallel operation
Electromagnetic interference	Fungus
Surge rating	
Ripple voltage	
Parallel operation	
Intermittent duty	
Transient load response	
Fan motor	
Endurance	
High temperature	
Sunshine	
Low temperature	
Humidity	
Sand and dust	
Salt fog	
Rain	
Air transportability	

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* 4.5 Quality conformance tests. The quality conformance tests shall consist of the individual tests.

4.5.1 Individual tests. Each power supply shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Dielectric

c. Operational

d. Regulation

e. Efficiency.

4.6 Test methods

4.6.1 Examination of product. The power supply shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, weight, and dimensions.

4.6.2 Dielectric. The power supply shall be subjected to a dielectric test of twice rated input voltage +1,000V rms, 60-cycle as primary-to-secondary and primary-to-ground, and 500V secondary-to-ground for 1 minute. The insulation shall not fail nor be damaged during the test. Upon completion of all environmental tests, this test shall be repeated using voltage values of 80 percent or above.

4.6.3 Operational. The power supply shall be operated for 1/2 hour at rated load conditions and followed immediately by operation at 150 percent rated current for 3 minutes. The power supply shall then be tested for normal operation of all components. Overload relays shall be bridged during the overload test. Immediately at the conclusion of this test, the power supply shall meet the regulation and efficiency requirements of the detail specification when tested as specified in 4.6.4 and 4.6.5 herein.

4.6.4 Regulation. The power supply shall be operated for 1/2 hour at rated conditions with the voltage maintained constant at the input terminals. Output voltage measurements shall be made while the output current is decreased from rated load to no load in steps not to exceed 5 percent of the rated current. This procedure shall also be followed while the output current is increased from no load to rated load. The total output voltage variation during this test shall not exceed the value specified in the detail specification.

4.6.5 Efficiency. The power supply shall be operated for 30 minutes at rated input voltage and rated output load. In the case of power supplies with dual input voltage provisions, the power supply shall be operated at the lower value of rated voltage for the purpose of accomplishing this test. The input and

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output power shall be determined at the input and output studs exclusive of all leads. The overall efficiency of the power supply at full load shall be not less than that specified in the detail specification.

4.6.6 Stability. The portable power supply shall be tested for compliance with the stability requirements specified in 3.5.7. The power supply shall be tilted 30° from the normal position and, upon release, shall return to its normal stationary position.

4.6.7 Electromagnetic interference. The power supply shall be tested for radiated and conducted interference in accordance with MIL-STD-461 and MIL-STD-462, interference control requirements, class Gp except the antenna for the radiated test shall be placed not more than 6 feet from the outer perimeter of the unit, at the point of maximum interference.

4.6.8 Surge rating. The power supply shall be tested for compliance with the surge rating requirements specified in 3.6.3. The power supply shall be subjected to 400 percent rated current for 3 seconds at three surges per minute for 30 minutes. Overload relays shall not be bridged during this test, and interruption of current shall be performed on the output side to permit uninterrupted operation of the cooling device. The overload relays shall not open during this test, and neither the power supply nor any component part shall be damaged.

4.6.9 Ripple voltage. Ripple voltage shall be determined by means of a peak-reading vacuum tube voltmeter in series with a 4.0 microfarad capacitor connected across the output terminals of the power supply while operating from no load to rated load. Peak voltage readings shall be taken with the voltmeter successively connected to each of the polarities, and the higher of the two readings shall not exceed the values specified in the detail specification.

4.6.10 Parallel operation. With the power supply voltage regulator operating and under load conditions varying from 20 to 100 percent of the combined rating of the connected power supplies or from 20 to 100 percent of the rated load, the actual kilowatt load of any power supply expressed as a percentage of its continuous kilowatt rating shall not differ by more than 10 percent from the total combined kilowatt load on the connected power supplies expressed as a percentage of the combined rating.

4.6.11 Intermittent duty. The power supply shall be operated at 125 percent rated load for 3 minutes followed by 2 minutes operation at rated load. This operating cycle shall be repeated six times. No injury to the power supply or component part shall result from this test.

4.6.12 Transient load response. The power supply shall be tested to determine compliance with the transient load response characteristics specified in 3.6.4. A transient load equal to the overload rating shall be applied to and removed from the power supply, thus causing the power supply output to change from no load to overload and vice versa. Values of output current and voltage shall be oscillographically recorded during the test.

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4.6.13 Fan motor. It shall be demonstrated that the power supply will stop operating when the fan motor is no longer energized.

4.6.14 Endurance. The power supply shall be subjected to a minimum of 200 hours operation at rated load. During this test, the transformer temperature rise shall not exceed 50°C for class A insulation, 75°C for class B insulation, nor 120°C for class H insulation when measured with thermocouple placed between the windings and the core. Temperature readings shall be recorded at least every 8 hours during the continuous run as well as immediately upon shutoff of the input and cooling device. Upon completion of this test, the power supply shall be tested to determine compliance with the efficiency and regulation requirements of the detail specification.

4.6.14.1 Test time. Intervals of endurance test time less than 1 hour terminated by a parts failure shall not be credited to the endurance test. Endurance test time shall be credited in 1/2-hour intervals, except where shorter periods are a test requirement.

4.6.15 Environmental tests. The power supply shall be subjected to the following tests in accordance with MIL-STD-810. There shall be no damage to excessive corrosion resulting from these tests, and the power supply unit shall meet the requirements for efficiency and regulation specified in the detail specification when tested in accordance with 4.6.4 and 4.6.5. Upon completion of all the environmental tests, the power supply shall be subjected to the dielectric test specified in 4.6.2.

a. High temperature - method 501, procedure I

b. Sunshine - method 505, procedure I

c. Low temperature - method 502, procedure I

d. Humidity - method 507, procedure I

e. Fungus - method 508, procedure I
f. Sand and dust - method 510,
f. Sand and dust - method 510, procedure I

g. Salt fog - method 509, procedure I

h. Rain - method 506, procedure I.

4.6.16 Air transportability. Unless otherwise specified, the power supply shall be tested for air transportability in accordance with MIL-A-8421.

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4.6.17 Reliability demonstration. Satisfactory reliability shall be considered demonstrated when both the following apply:

a. Prediction - A reliability prediction, using data, estimates, modifying factors, and procedures acceptable to the procuring activity shall yield a MTBF prediction not less than the requirements of 3.5.3

b. All tests specified herein shall be completed satisfactorily with no failure that could prevent safe or satisfactory performance of primary functions. Any test resulting in a failure shall be documented. Apparent failures that are caused by some unintended factor external to the equipment (e.g. test equipment failure) need not be counted as reliability failures if the causes can be substantiated to the satisfaction of the procuring activity. Acceptance for reliability purposes shall not be made until, to the satisfaction of the procuring activity, each failure has been analyzed, the cause determined, design or other modifications made to preclude repetition of the failure or of related failures, the failed equipment has been successfully retested with the modifications included, and action has been taken to reflect the modifications in equipment to be delivered. Any retest shall include all tests, in addition to a particular failed test, that could be influenced by the modifications.

* 4.6.18 Maintainability demonstration. All maintenance tasks other than depot maintenance shall be performed. Failures need not be induced for corrective tasks, but each task shall be accomplished using the fault isolation procedures and the test and calibration equipment and tools that are or are planned to be included in the applicable maintenance manuals as though a failure existed. The active maintenance downtime, the tools and equipment required, and any obstructions or conditions hindering the task or presenting potential hazard to parts or to personnel shall be observed and recorded. Signs or excessive wear or other indication of potential failure shall be recorded and the information supplied to the design activity for study and for recommendation of any appropriate modifications to be reflected in future items delivered. Such recommendations shall be subject to the approval of the procuring activity before being implemented or successful repeat may be required of other tests specified herein that may be influenced by the modifications. The active maintenance downtime need not include administrative time (such as completing records) or supply time (such as obtaining parts). It shall be demonstrated that the item is in satisfactory operating condition after all maintenance tasks have been completed. Maintenance tasks to be listed and accomplished need not be carried down to the level of separate tasks for bolts, nuts, and other hardware that are not vital to satisfactory performance and that would normally be removed and replaced as part of other maintenance.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

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* 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified in the detail specification, selection of the appropriate method shall be accomplished using the preservation selection tables of MIL-STD-794 and applied in accordance with MIL-P-116.

5.1.2 Level C. The power supply shall be preserved and packaged in a manner that will provide protection against corrosion, deterioration, or physical damage so that serviceability is assured during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Power supply, preserved and packaged as specified in 5.1.1, shall be packed in appropriate exterior container, overseas type, selected from table I of MIL-STD-794 and the packing procedures of that standard shall apply.

5.2.2 Level B. Power supply, preserved and packaged as specified in 5.1.1 shall be packed in appropriate exterior container, domestic type, selected from table I of MIL-STD-794 and the packing procedures of that standard shall apply.

5.2.3 Level C. Power supply shall be packed in exterior type shipping container in a manner that will insure safe transportation at the lowest rate to the point of delivery and shall meet, as a minimum, the requirements of carrier rules and regulations.

5.3 Marking. In addition to any marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The power supply covered by this specification is intended for use as a ground supply of direct-current electrical power.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type of input cable required (see 3.7.3.1)

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c. Location and conditions for preproduction testing (see 4.4)

d. Selection of applicable levels of preservation, packaging, and packing (see 5.1).

6.3 Definition. For the purpose of this specification, the following definition will apply:

6.3.1 Intermittent duty. Intermittent duty will consist of 5-minute intervals of operation, 3 minutes of which shall be at 125-percent full load and 2 minutes at 100-percent full load.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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